

Blockchain Lab

EXP 8

Aim - To implement the Blockchain platform Ganache and deploy a simple Solidity smart contract using Metamask and Remix IDE.

Theory - Ganache is a personal Ethereum blockchain used for testing and development. It allows developers to deploy contracts, develop decentralized applications (DApps), and run tests in a safe and deterministic environment.

Ganache provides:

- Instant mining of blocks
- Pre-funded test accounts
- Transaction history view
- Graphical interface to monitor blockchain activities

It is part of the Truffle Suite and is commonly used with Remix and Metamask for smart contract deployment.

Steps to Connect Ganache with Metamask and Remix IDE:

1. Install Ganache:
 - Download and install Ganache from the Truffle Suite website.
 - Launch it to start a local blockchain network with test Ether accounts.
2. Connect Ganache Accounts with Metamask:
 - Open Metamask and switch to a Custom RPC Network.
 - Enter the Ganache RPC server URL (e.g., [HTTP://127.0.0.1:7545](http://127.0.0.1:7545)).
 - Import private keys of Ganache accounts into Metamask.
3. Connect Remix IDE with Metamask:
 - Open Remix IDE in the browser.
 - Choose Injected Web3 as the environment in the “Deploy & Run Transactions” tab.
 - Remix will automatically connect to the Metamask account (linked with Ganache).
4. Create a Simple Solidity Smart Contract:
 - Write a Solidity program related to your mini project (e.g., Student Record, Donation Tracker, etc.).

```
pragma solidity ^0.8.0;
contract SimpleStorage {
    uint public data;
    function set(uint x) public {
        data = x;
    }
}
```

5. Compile and Deploy the Contract:
Compile using Remix.
 - Deploy via the Metamask account connected to Ganache.
 - Approve the transaction in Metamask.
6. Check Transaction Details on Ganache:
 - View mined transactions, block details, and balances in the Ganache interface.
7. Interact with the Smart Contract:
 - Use Remix to call contract functions (`set()` or `get()`).
 - Observe the state changes and corresponding transactions on Ganache.

Code -

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

contract Escrow {

    address public buyer;
    address payable public seller;
    address public arbiter;

    bool public buyerApproved;
    bool public sellerApproved;

    constructor(address payable _seller, address _arbiter) payable {
        buyer = msg.sender;
        seller = _seller;
        arbiter = _arbiter;
    }

    function approve() public {
        require(msg.sender == buyer || msg.sender == seller, "Not authorized");

        if (msg.sender == buyer) {
            buyerApproved = true;
        }

        if (msg.sender == seller) {
            sellerApproved = true;
        }

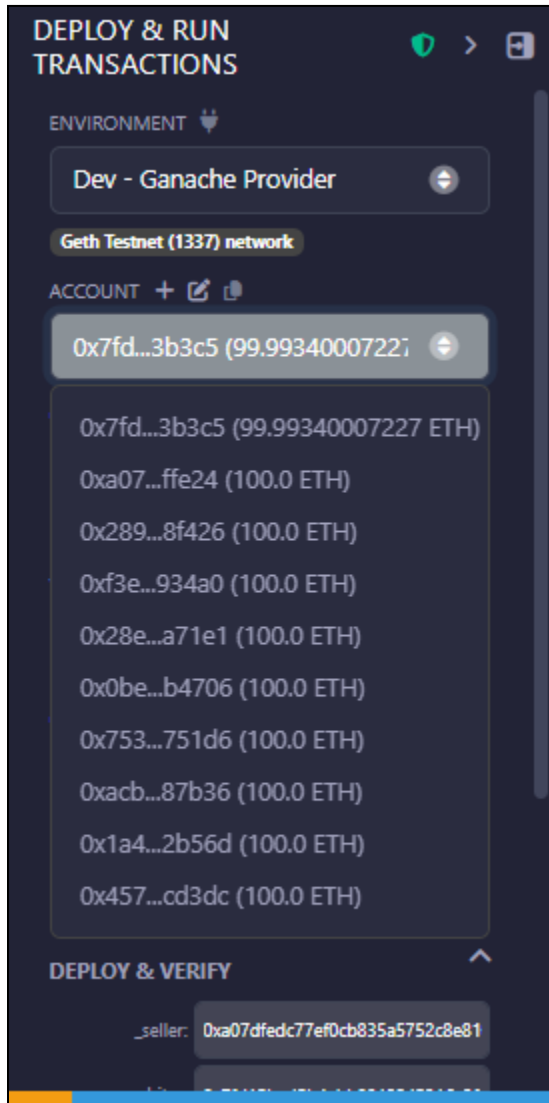
        if (buyerApproved && sellerApproved) {

            uint amount = address(this).balance;

            (bool success, ) = seller.call{value: amount}("");
```

```
        require(success, "Transfer failed");
    }
}
```

Output -





Add wallet

Private key

Imported private keys are backed up to your account and sync automatically when you sign in with the same Google or Apple login.
[Learn more](#) about how imported keys work

Select type

Private key ▾

Enter your private key string here:

.....|



Cancel

Import

ACCOUNT INFORMATION

ACCOUNT ADDRESS

0xA07DFedc77EF0cB835a5752c8e8108E7cA4ffe24

PRIVATE KEY

0xfeb942cbaeb4cc850a33f5c87a3e5af51e76d998538bff2fad3f0c643128ed24

Do not use this private key on a public blockchain; use it for development purposes only!

DONE

ACCOUNTS

BLOCKS

TRANSACTIONS

CONTRACTS

EVENTS

LOGS

SEARCH FOR BLOCK NUMBERS OR TX HASHES

CURRENT BLOCK
2

GAS PRICE
20000000000

GAS LIMIT
6721975

HARDFORK
MERGE

NETWORK ID
5777

RPC SERVER
HTTP://127.0.0.1:7545

MINING STATUS
AUTOMINING

WORKSPACE
QUICKSTART

SAVE

SWITCH

BLOCK
2

MINED ON
2026-03-17 11:46:30

GAS USED
3000000

1 TRANSACTION

BLOCK
1

MINED ON
2026-03-17 11:46:19

GAS USED
3000000

1 TRANSACTION

BLOCK
0

MINED ON
2026-03-17 11:36:01

GAS USED
0

NO TRANSACTIONS

CURRENT BLOCK
2

GAS PRICE
2000000000

GAS LIMIT
6721975

HARDFORK
MERGE

NETWORK ID
5777

RPC SERVER
HTTP://127.0.0.1:7545

MINING STATUS
AUTOMINING

WORKSPACE
QUICKSTART

SAVE

SWITCH

TX HASH

0xb85506ee1d0385b8a59d33a0e97cbdc501667e7d45ef7ac1f5beec6aa9a79243

CONTRACT CREATION

FROM ADDRESS

0x7fd15bAd5b4AbB0912345316A3fcD453B2a3b3C5

CREATED CONTRACT ADDRESS

0x7b10fEE11188F75915ce12A07120d8442A7638af

GAS USED

3000000

VALUE

0

TX HASH

0xa4f74fed3fc1e12bcc5777e0337466b545001a6ecac86fdea5c9e04f4f7aa085

CONTRACT CREATION

FROM ADDRESS

0x7fd15bAd5b4AbB0912345316A3fcD453B2a3b3C5

CREATED CONTRACT ADDRESS

0x018b4fb3336095E9cdf9988B0d8f8C7D22e24ECC

GAS USED

3000000

VALUE

0

Conclusion - Solidity smart contract were deployed on remix IDE and transaction were executed successfully on ganache blockchain with the help of metamask wallets.